Problem Sheet 4

discussion: Tuesday, 03.11.2026

4.1. Show that, for $r \geq p$, one has $S^{p,r}(\Delta) = \mathcal{P}_p$ irrespective of the partition Δ.

Also show that the not-a-knot cubic spline interpolating four data points reproduces the cubic polynomial through these points.

4.2. Consider the 2π-periodic functions

$$f(x) = \frac{1}{2 - \cos(x)}, \qquad g(x) = |\sin(x/2)|.$$

For $n = 2m + 1$, use the knots $x_j = 2\pi j/n$, $j = 0, \ldots, n-1$.

a) Write a `matlab` or `python` routine that computes and evaluates the classical trigonometric interpolant

$$q_n(x) = \sum_{k=-m}^{m} d_k e^{\mathbf{i}kx}$$

using one FFT of the sampled data. Specify which entries of $\widehat{\mathbf{y}}/n$ give the coefficients d_k for negative k. Check that your routine interpolates the data and gives real values up to rounding errors.

b) For $n \in \{9, 17, 33, 65, 129\}$, approximate $\|f - q_n\|_{\infty,[0,2\pi]}$ and $\|g - q_n\|_{\infty,[0,2\pi]}$ by sampling at 10001 uniformly distributed evaluation points. Plot each error versus n using both `semilogy` and `loglog`. Which plot suggests exponential convergence, and which suggests algebraic convergence? Relate the results to the smoothness of the periodic functions.

4.3. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be given by

$$\mathbf{A} = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & & -1 \\ -1 & 2 & -1 & & & \\ & -1 & 2 & -1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -1 & 2 & -1 \\ -1 & & & & -1 & 2 \end{pmatrix} + \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix} \qquad \text{with } h = 1/N.$$

The matrix $\mathbf{A}$ is a circulant matrix (compare Example 1.60 in the lecture notes). Read Example 1.59 in the lecture notes on how linear systems with circulant matrices can be solved using FFT. For $\mathbf{b} = (1, \ldots, 1)^T$, write a code that implements this solution procedure for the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Use different values of $N = 2^j$, $j \geq 2$ and plot the computational times of computing the corresponding solution vectors $\mathbf{x} \in \mathbb{R}^N$ versus N.

4.4. (Fast polynomial multiplication) Let

$$A(z) = 1 + 2z - z^2, \qquad B(z) = 2 - z + z^3,$$

with coefficient vectors $\mathbf{a} = (1, 2, -1)^\top$ and $\mathbf{b} = (2, -1, 0, 1)^\top$ in the monomial basis. Compute the coefficients of $A \cdot B$ directly.

a) For n-periodic sequences $\mathbf{u}, \mathbf{v}$ prove the convolution theorem $F_n(\mathbf{u} * \mathbf{v}) = (F_n\mathbf{u}) \cdot (F_n\mathbf{v})$, where $\cdot$ denotes componentwise multiplication.

b) Extend both coefficient vectors $\mathbf{a}, \mathbf{b}$ with zeros to length $n = 8$ and calculate the product coefficients using two FFTs and one inverse FFT.

4.5. (Frequency analysis and filtering) A periodic signal is sampled at $x_j = 2\pi j/n$, $j = 0, \dots, n-1$:

$$y_j = s(x_j) + \eta(x_j), \qquad s(x) = \sin(3x) + \frac{1}{2}\cos(7x), \qquad \eta(x) = \frac{3}{10}\sin(25x) + \frac{1}{5}\cos(40x).$$

a) For $n = 128$, plot the samples and the amplitude spectrum $|\widehat{y}_k|/n$. Label the frequency axis by the signed integers $0, \dots, n/2, -n/2+1, \dots, -1$ and identify all nonzero modes.

b) Apply a low-pass filter: keep only coefficients with frequencies $|k| \le 10$, and set all others to zero. Use the inverse FFT to obtain $\widetilde{y}_j$ and plot the filtered samples together with s. Derive the discrete Parseval identity

$$\sum_{j=0}^{n-1} |v_j|^2 = \frac{1}{n} \sum_{k=0}^{n-1} |(F_n \mathbf{v})_k|^2$$

from Theorem 1.45. Use it to predict the root mean square error $\big(n^{-1} \sum_j |y_j - s(x_j)|^2\big)^{1/2}$ before filtering, and compare with the errors before and after filtering computed from the samples.

c) Undersampling: Repeat the above example with $n = 32$. Does aliasing (compare Exercise 3.5) happen?